# D 133 | Re-derive, do not copy

PRIVATE FICTIONAL integrated review. No competitor contact, coordination, campaign, vendor switch, budget, private data, account change or spend.

| | |
|---|---|
| **Changed game** | Own crore/quarter utilities: T 70, R 40, P 15, S 10. Strict D dominance; mutual D stage Nash. Infinite common discount .5, perfect public observation and credible grim: C 80 versus deviation 85. Weak threshold 6/11. Change ONLY T 50: deviation 65, threshold 2/7. This is parameter sensitivity, not a regulation or real-company finding. |
| **Weak boundary and horizon** | At either threshold, tested values tie. Infinite sum starts with current payoff; normalized values multiply by 1-d. Known finite N with unique stage Nash and no terminal rewards gives all D, not C until last period. Changed payoffs may leave strict-PD scope; do not reuse verdict. |
| **Separate noise toy** | Independent public false-alarm probability .02 per still-cooperative period, N 10. At least one 1-.98^10. Policy evaluation assumes current reward first, then permanent punishment NEXT period. It supplies no signal law for true deviations and no equilibrium/optimal-forgiveness verdict. |
| **Fresh finance units** | 100000 impressions, CPM 250, CTR 2%, 5% purchases/click gives spend 25000, 100 distinct new customers, CAC 250. CPM 300 alone gives 300. Separate 60% customers at 300/40%at 200 gives 260. Monthly cash lakh 12+4-2-3-1-.5=9.5, delta-2.5; net variable product/fulfilment already deducted from 4. |
| **Evidence is not a slogan** | Unknown rarity/imitability remain unknown; organization means benefit capture, not ownership. New category demand differs from own/rival switching. Network value differs from volume/scale; rail roles differ from fee applicability. Foothold/performance differs from adoption/displacement. Scenario/quantity trigger differs from authority. |
| **Actual work remains blank** | Record independently attempted correction, source objection and delayed transfer only if actually performed. Authored tests do not establish learner mastery, empirical business truth, full source/AV, native parity, account/legal clearance or overall 10. |

# Inputs | Utility, periods and assumptions

Synthetic symmetric payoffs. Discount is a per-period factor, not an annual interest rate. False alarm is ONE public event per cooperative period, not one independent event per player.

| | |
|---|---|
| **Temptation T, crore/quarter utility** | 70.0000 |
| **Reward R, crore/quarter utility** | 40.0000 |
| **Punishment P, crore/quarter utility** | 15.0000 |
| **Sucker S, crore/quarter utility** | 10.0000 |
| **Common per-period discount delta** | 0.5000 |
| **Public false-alarm probability** | 0.0200 |
| **Known finite periods / alarm window** | 10.0000 |
| **Changed temptation only, same units** | 50.0000 |

| | |
|---|---|
| **Payoff entries valid numbers?** | 1.000000 |
| **Discount valid, 0 <= delta < 1?** | 1.000000 |
| **Noise/window valid?** | 1.000000 |
| **Changed temptation valid?** | 1.000000 |

# Inputs | Utility, periods and assumptions

Synthetic symmetric payoffs. Discount is a per-period factor, not an annual interest rate. False alarm is ONE public event per cooperative period, not one independent event per player.

| | |
|---|---|
| **Campaign impressions** | 100000.0000 |
| **Base CPM, INR/1000impressions** | 250.0000 |
| **Changed CPM, same rates** | 300.0000 |
| **Click-through fraction** | 0.0200 |
| **Purchase per click fraction** | 0.0500 |

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|---|---|
| **Customer share at channelA CAC** | 0.6000 |
| **ChannelA CAC, INR/customer** | 300.0000 |
| **ChannelB CAC, INR/customer** | 200.0000 |

| | |
|---|---|
| **Opening cash, lakh INR** | 12.0000 |
| **Collected net of paid variable costs, lakh** | 4.0000 |
| **Marketing cash, lakh** | 2.0000 |
| **Fixed cash, lakh** | 3.0000 |
| **Additional working capital, lakh** | 1.0000 |
| **Principal payment, lakh** | 0.5000 |

| | |
|---|---|
| **Acquisition inputs valid?** | 1.000000 |
| **Customer mix valid?** | 1.000000 |
| **Cash inputs valid?** | 1.000000 |

Negative stage utility and net collected flow allowed. Discount 0<=d<1; integer N 1..1000 is test bound, not theorem. Input validity does not verify empirical assumptions. Models remain separate; static prose is baseline only after edits.

# Stage | Best responses before repetition

Symmetric stage game. D against C gives T; D against D gives P. Strict PD order T>R>P>S is a scope gate, not a claim about real markets.

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|---|---|
| **Gain from D rather than C against C** | 30.000000 |
| **Gain from D rather than C against D** | 5.000000 |
| **Strict prisoner-dilemma order? 1=yes** | 1.000000 |

| | |
|---|---|
| **D strict best response against either? 1=yes** | 1.000000 |
| **Each-player gain: mutual C versus mutual D** | 25.000000 |

Baseline differences 30, 5; strict PD 1 and dominant D 1; mutual C raises each payoff 25. If payoff order changes, generic arithmetic remains visible but the PD-specific grim/finite conclusions are held. MutualD Nash is not Pareto efficiency.

# Infinite | Perfect-observation grim test

Fixed symmetric game, infinite horizon, common delta, perfect public observation. Unnormalized discounted sums starting with current period. MutualD is credible because D best responds to D.

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|---|---|
| **Always-C path present value** | 80.000000 |
| **One deviation now then perpetual mutual D** | 85.000000 |
| **Cooperation minus deviation value** | -5.000000 |

| | |
|---|---|
| **Weak incentive threshold, strict PD only** | 0.545455 |
| **Normalized cooperation value** | 40.000000 |
| **Normalized one-deviation value** | 42.500000 |

| | |
|---|---|
| **Conditional model verdict** | IC FAILS |

Baseline 80 vs 85, margin-5, threshold 6/11, normalized 40 vs 42.5. Delta.5 fails; 6/11 is weak equality, not strict preference. MutualD remains another equilibrium. This does not prove real collusion/cooperation, select a unique equilibrium or establish noisy-monitoring IC.

# Finite | Known final period

Only under complete information, fixed stage game with unique mutual D stage Nash, no terminal reward and known N. AllD by backward induction; stage-order gate prevents silently using it in another game.

| | |
|---|---|
| **Discounted N-period annuity starting now** | 1.998047 |
| **AllD finite value under scoped model** | 29.970703 |
| **AllC arithmetic comparator, NOT equilibrium** | 79.921875 |
| **Last-period gain from D against C** | 30.000000 |
| **Backward-induction scope status** | KNOWN N: ALL D |

Ten-period all D value 29.970703125 atdelta.5; all C comparator 79.921875. Cooperative comparator is not predicted finite equilibrium. Unknown horizon, incomplete information or multiple terminal equilibria require another model; do not generalize all D to all repeated interaction.

# Noise | False alarms are not deviations

Separate policy model. One public false alarm with probability p per still-cooperative period, independent over time. Payoff R occurs before alarm, then punishment P begins next period forever. No true-deviation signal law supplied.

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|---|---|
| **No false alarm in N independent opportunities** | 0.817073 |
| **At least one false alarm in N opportunities** | 0.182927 |

| | |
|---|---|
| **Perpetual punishment-state value** | 30.000000 |
| **Cooperative-state POLICY value, not IC** | 79.019608 |
| **Bellman identity residual** | 0.000000 |
| **Loss versus perfect always-C path** | 0.980392 |

| | |
|---|---|
| **Equilibrium/forgiveness verdict** | POLICY ONLY: IC UNKNOWN |

At least one false alarm 1-.98^10=18.292719311%; not 20% except a first-order approximation. Expected policy value is not an optimal policy or equilibrium. Forgiveness may reduce false punishment but also weaken deterrence; need deviation detection, public/private signals, beliefs and full strategy tests. No optimal forgiveness claim.

# Shock | Change temptation, not the whole game

Same R 40, P 15, S 10 and delta.5; only T changes 70->50. Arithmetic remains visible; strict-PD scope prevents copying threshold conclusions into another ordering.

| | |
|---|---|
| **Changed strict-PD order?1=yes** | 1.000000 |
| **Changed one-deviation value** | 65.000000 |
| **Cooperation minus changed deviation** | 15.000000 |
| **Changed weak threshold** | 0.285714 |

| | |
|---|---|
| **Changed conditional verdict** | WEAK IC HOLDS |

Atdelta.5: 80 vs 65, margin 15; threshold 10/35=2/7. At 6/11 original tested values tie; at 2/7 changed values tie. Equality is weak incentive, not unique equilibrium. No real regulation/payoff estimate or competitor action.

# Acquisition | Fresh campaign denominator

INR/campaign and distinct new customers. Rates held fixed under CPM-only change; no causal conversion or attribution evidence.

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|---|---|
| **Base spend, INR** | 25000.000000 |
| **Clicks** | 2000.000000 |
| **New customers** | 100.000000 |
| **Base CAC, INR/customer** | 250.000000 |

| | |
|---|---|
| **Changed spend, INR** | 30000.000000 |
| **Changed CAC, same rates** | 300.000000 |

25000/100=250; 30000/100=300. Zero modeled customers makes CAC undefined, not zero. Expected counts may be fractional after edits. Distinct/new assumptions require actual reconciliation before using real customer data. No campaign executed.

# Blend | Separate cohort, customer weights

60% acquired customers at 300 and 40%at 200, not campaign CPM or spend shares. Independent of the previous campaign model.

| | |
|---|---|
| **Customer-weighted CAC, INR/customer** | 260.000000 |

Baseline .6*300+.4*200=260. Unweighted 250 or reversed-weight 240 is wrong. If the shares were SPEND shares, the harmonic calculation would be different. No actual cohort or incremental acquisition established.

# Cash | Endpoint, not a profit statement

Lakh INR/month. Collected sales less paid variable product/fulfilment costs is 4, before listed marketing/fixed/additional working-capital/principal. No other flows/tax assumed.

| | |
|---|---|
| **Net monthly cash change** | -2.500000 |
| **Closing cash endpoint** | 9.500000 |
| **Endpoint decline** | 2.500000 |

12+4-2-3-1-.5=9.5, net change-2.5. Do not subtract paid product/fulfilment twice; additional working capital not already deducted from 4. Principal is cash but not operating expense. Endpoint is not profit, intramonth minimum, runway or funding permission.

# Review | Which evidence changes the verdict?

Each item is a reasoning check, not an assessment of an actual company. Source facts and toy calculations are separate.

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|---|---|
| **VRIO** | Driver-network value does not establish rarity or costly imitation. Leave unknowns open; organization asks whether rights, processes and incentives capture benefit. Ownership alone does not answer it. Matched rivals/time/geography and counterevidence are needed. |
| **New category demand** | Prior nonconsumers, own-product diversion and rival-product switching are distinct categories. Record time, denominator, previous alternative, paid demand and contribution. A different target value curve establishes neither willingness to pay nor cost feasibility. |
| **Platform and rail** | More volume during subsidy is not evidence that participation itself adds value. Merchant participation helping payers is cross-side; processing scale is different. Interoperability can reduce reach exclusivity without equal app service/trust. NPCI roles page is not operative fee/cap verification. |
| **Disruption** | Cheap/digital/successful entry is insufficient. Serving incumbent best customers first suggests sustaining competition rather than proven low-end/new-market foothold. Meeting a performance threshold does not prove adoption or displacement. Track viable economics and later outcomes. |
| **Resilience loop** | Service failures->refunds/backlog->cash pressure->capacity cuts->failures closes four positive hypothesized links, with delays. Two negative links result if using capacity LEVEL instead of CUTS, still reinforcing. Polarity alone predicts no calibrated path. |
| **Trigger and authority** | Example 7 day service-failure rate>5% on>=100 eligible orders, review within 1 day, fictional operations role then finance/sponsor escalation. Need source/window/false alarms/fallback, actual accepted owner, contract/data/account/budget authority. Trigger satisfaction is not approval. |

# Evidence | Coverage and disagreement travel with claims

13 November 2026. Three repeated-game sources revisited in selected written sections; prior strategy/resilience source evidence reused, not reread or expanded by repetition.

| | |
|---|---|
| **Perfect and finite games** | https://web.stanford.edu/~jdlevin/Econ%20286/Repeated%20Games%20I.pdf selected introduction/grim proof, not all later proofs. Its 1, 2, 0, -1 table differs. https://pages.nyu.edu/debraj/Courses/05UGGameLSE/Handouts/05uggl02.pdf selected setup/finite theorem/multiple-equilibrium example/grim, not entire proof. Its 2, 3, 1, 0 table differs. Neither supplies our 40/70/15/10. |
| **Noisy monitoring** | https://ocw.mit.edu/courses/14.126-game-theory-spring-2024/mit14_126_s24_lecture_5-6_repeated_games.pdf selected tolerant-grim/SSE efficiency remarks and monitoring definitions. Source distinguishes optimality WITH specified signal/IC/class restrictions. Our false-alarm policy lacks true-deviation law and cannot borrow that optimality claim. Not full proofs/figures/course. |
| **Role is not fee evidence** | https://www.npci.org.in/product/upi/roles-responsibilities prior selected roles evidence only; operative current fees/cap applicability remain unresolved from D 130. No fee calculator or legal conclusion. No real app-value-capture inference from rail participation. |
| **Theory and criticism** | https://www.christenseninstitute.org/theory/disruptive-innovation/ prior written theory; D 131 dated taxi/black-car disagreement and expert-case criticism remain. This review supplies no current company label, independent replication or predictive certainty. |
| **Continuity and scenarios** | https://www.ready.gov/business-continuity-plan prior written landing updated 25 March 2026; Planning Suite unsupported/unavailable, not manuals/videos. https://www.mckinsey.com/capabilities/strategy-and-corporate-finance/our-insights/the-use-and-abuse-of-scenarios prior written 2009 article; scenario heuristics not probabilities. D 132 OECD nonprediction/selected MIT loop definitions retained, not full reports. |
| **Limits** | Seven listed URLs are not independent business-efficacy replications. No source assets redistributed. Authored LibreOffice formulas only, not native Excel/Sheets parity, actual learner transfer, full AV/interactive source coverage, account/legal clearance or overall 10. No real coordination, pricing, campaign, vendor/data/budget/account action. |

# Actual work | No real rivalry or learner inference

Only record performed permission-cleared work. Unknown is not zero or confirmed mastery.

| Date/version | Threshold correction | Horizon/noise objection | Customer/cash units | Source objection | Reviewer finding | Delayed transfer |
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Actual work remains blank. Record only what occurred: dates, reasoning, error/source/correction, strongest objection and a new delayed transfer. No invented learner results, sponsor acceptance, actual budget release or signed rights.

# Failure log | Authored execution only

Observed native tests are added only after execution. Not actual learner mistakes, independent human audit, Excel/Sheets parity, real authority or overall 10/10.

| Controlled test | Observed authored outcome |
|---|---|
| **Independent baseline formulas** | PASS:46 baseline formulas independently recomputed. |
| **Changed game/discount/temptation** | PASS:10 changed payoff/discount/temptation sets. |
| **Finite and false-alarm bounds** | PASS:4 changed finite/false-alarm sets; policy only. |
| **Changed CAC/customer shares** | PASS:7 changed campaign/customer-share sets. |
| **Changed monthly cash** | PASS:3 changed monthly cash sets. |
| **Invalid controls and isolation** | PASS:16 invalid controls with independent model isolation. |
| **Broken-reference explicit holds** | PASS:6 broken references cause explicit holds/errors. |
| **Restored baseline/actual blank** | PASS:24 changed sets+restore;actual work blank. |

Only the original synthetic course exercise at https://top1os.dev/ supplies the case figures. No new empirical reading, real participants, company records, Indian legal clearance, actual commitments or source completeness claimed.